B4 Sub-Atomic Physics — Model Solutions, 2016

Question 1 — Neutron structure, decay and kinematics

Problem. Quark composition, spin, multiplet count of the neutron; constituent-quark estimate of m_n from a uniform sphere of radius $R = 1.5\,\text{fm}$; neutron β-decay diagram; absence of proton decay; minimum and maximum electron energies in $n \to p\, e^- \bar{\nu}_e$; conversion of the free-neutron lifetime to a width.

Quark content / spin / multiplet

Neutron $= udd$, spin $J = 1/2$, in the $J^P = 1/2^+$ baryon octet:

$$\boxed{udd, \quad J = \tfrac{1}{2}, \quad \text{8 hadrons in the multiplet}}$$

(a) RMS radial position

Uniform sphere, radius $R = 1.5\,\text{fm}$, density $\rho(r) = 3/(4\pi R^3)$. Mean-square radius:

$$\langle r^2 \rangle = \int_0^R r^2 \, \frac{4\pi r^2}{(4/3)\pi R^3} \, \mathrm{d}r.$$

Evaluate the integral:

$$\langle r^2 \rangle = \frac{3}{R^3} \int_0^R r^4 \, \mathrm{d}r = \frac{3R^2}{5}.$$

Take square root:

$$\Delta r = r_{\text{rms}} = R\sqrt{3/5}.$$

Insert $R = 1.5\,\text{fm}$:

$$\boxed{\Delta r = 1.5\sqrt{0.6}\,\text{fm} = 1.16\,\text{fm}}$$

(b) Quark momentum and energy

Heisenberg estimate, identifying Δp with the mean momentum:

$$\Delta p \sim \frac{\hbar}{\Delta r}.$$

Multiply by c, using $\hbar c = 197.3\,\text{MeV fm}$:

$$\Delta p\, c = \frac{197.3\,\text{MeV fm}}{1.16\,\text{fm}} = 169.8\,\text{MeV}.$$

Relativistic energy with $m_q c^2 = 5\,\text{MeV}$:

$$E_q = \sqrt{(\Delta p\, c)^2 + (m_q c^2)^2}.$$

Substitute numbers:

$$E_q = \sqrt{169.8^2 + 5^2}\,\mathrm{MeV} = \sqrt{28857}\,\mathrm{MeV}.$$

Evaluate:

$$\boxed{E_q \approx 169.9\,\mathrm{MeV}}$$

(c) Neutron-mass estimate and ratio

Sum over the three constituent quarks:

$$m_n c^2 \approx 3\,E_q = 3 \times 169.9\,\mathrm{MeV}.$$

Evaluate:

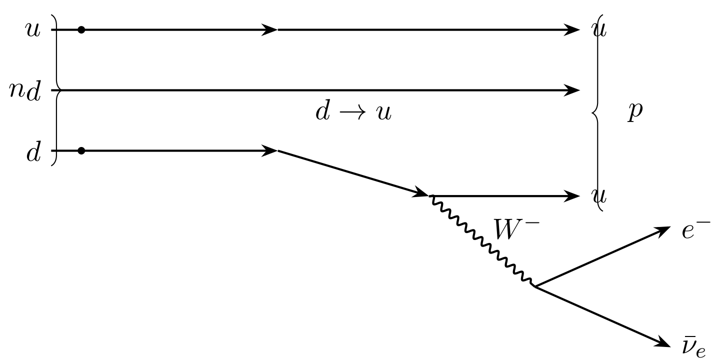
$$\boxed{m_n^{\mathrm{est}} \approx 510\,\mathrm{MeV}/c^2}$$

Ratio to the true neutron mass:

$$\frac{940}{510} \approx 1.84.$$

Improvements: include strong-interaction binding (gluon field energy, QCD potential), constituent-quark effective mass ($\sim 300\,\mathrm{MeV}$ rather than current $5\,\mathrm{MeV}$), and the colour-spin (hyperfine) interaction.

Neutron decay diagram



No proton decay. The proton is the lightest baryon with $B = 1$. Energy conservation requires

$$m_p > m_n + m_{e^+} + m_\nu \quad \text{for } p \to n e^+ \nu_e,$$

but $m_p < m_n$, so the analogous decay is kinematically forbidden.

Electron energy extrema in $n \to p\,e^- \bar{\nu}_e$

Three-body decay in the neutron rest frame; let p_e, p_p, p_ν denote three-momenta with $\mathbf{p}_p + \mathbf{p}_e + \mathbf{p}_\nu = 0$.

Minimum electron energy: the electron is at rest, $\mathbf{p}_e = 0$. The proton and (massless) antineutrino balance momenta:

$$\mathbf{p}_p = -\mathbf{p}_\nu, \qquad |\mathbf{p}_p| = |\mathbf{p}_\nu|.$$

So the minimum is just the rest energy:

$$\boxed{E_e^{\min} = m_e c^2}$$

Maximum electron energy: the antineutrino has zero momentum (massless $\Rightarrow E_\nu = 0$). Electron and proton recoil back-to-back:

$$\mathbf{p}_p = -\mathbf{p}_e, \qquad |\mathbf{p}_p| = |\mathbf{p}_e| \equiv p.$$

Four-momentum conservation: $E_e + E_p = m_n c^2$. Square the proton four-momentum:

$$(m_n c^2 - E_e)^2 = (pc)^2 + (m_p c^2)^2.$$

Use $(pc)^2 = E_e^2 - (m_e c^2)^2$:

$$m_n^2 c^4 - 2m_n c^2 E_e + E_e^2 = E_e^2 - m_e^2 c^4 + m_p^2 c^4.$$

Cancel E_e^2:

$$m_n^2 c^4 - 2m_n c^2 E_e = m_p^2 c^4 - m_e^2 c^4.$$

Solve for E_e:

$$\boxed{E_e^{\max} = \frac{m_n^2 + m_e^2 - m_p^2}{2\,m_n}\,c^2}$$

Decay width from lifetime

Use $\Gamma = \hbar/\tau$ with $\tau = 15 \times 60\,\mathrm{s} = 900\,\mathrm{s}$:

$$\Gamma = \frac{\hbar}{\tau} = \frac{6.582 \times 10^{-16}\,\mathrm{eV\,s}}{900\,\mathrm{s}}.$$

Evaluate:

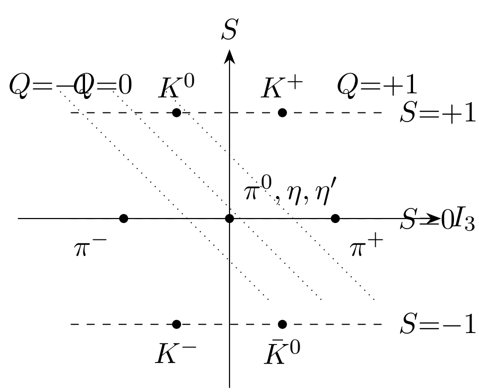
$$\boxed{\Gamma \approx 7.3 \times 10^{-19}\,\mathrm{eV}}$$

Question 2 — Light mesons and resonant production

Problem. Pseudoscalar nonet of K and π; Feynman diagrams and suppression factors for $\pi^+ \to \mu^+\nu_\mu$, $K^+ \to \mu^+\nu_\mu$, $K^+ \to \pi^+\pi^0$; resonance cross-section ratio for ρ^0 vs K^0 produced in $\pi^+\pi^-$ at $E = 249$ and $250\,\mathrm{MeV}$ per beam; pion radius of curvature in a $1\,\mathrm{T}$ field.

Pseudoscalar meson nonet

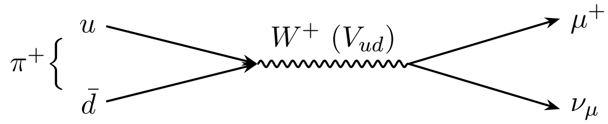
Axes: third isospin component I_3 (charge $Q = I_3 + Y/2$) and strangeness S.



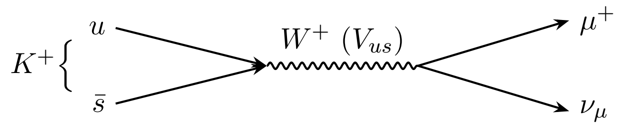
Nine states: triplet $\pi^{\pm,0}$ and singlets η, η' at $S = 0$; (K^+, K^0) at $S = +1$; $(K^-, \bar{K}^0)$ at $S = -1$.

Feynman diagrams and suppression factors

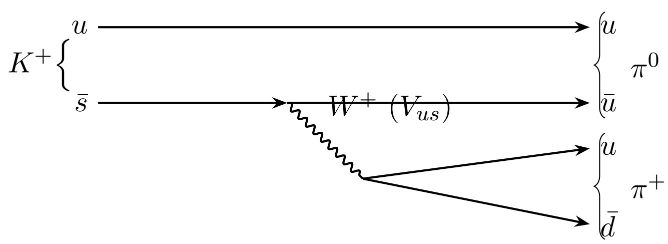
$\pi^+ \to \mu^+\nu_\mu$ ($\bar{d}u$ annihilation, V_{ud}):



$K^+ \to \mu^+\nu_\mu$ ($\bar{s}u$ annihilation, V_{us}):



$K^+ \to \pi^+\pi^0$ (spectator u; $s \to u$ via $W^+ \to u\bar{d}$, V_{us}):



Suppression factors.

- $\pi^+ \to \mu^+\nu$ vs. $K^+ \to \mu^+\nu$: both share the same leptonic helicity factor $(m_\mu/m_P)^2(1-m_\mu^2/m_P^2)^2$ and a CKM factor (V_{ud}^2 vs V_{us}^2). The pion partial width is suppressed by the much smaller phase space available, $\Gamma \propto m_P\,(m_\mu/m_P)^2(1-m_\mu^2/m_P^2)^2$, which is dominated by m_P. So pion-decay is *phase-space suppressed* relative to kaon-decay.

- K^+ decays vs. $\pi^+ \to \mu^+\nu$: kaon decays are Cabibbo-suppressed by $|V_{us}|^2 = \sin^2\theta_C \approx 0.05$ (vs. $|V_{ud}|^2 \approx \cos^2\theta_C$). $K^+ \to \mu^+\nu$ is additionally helicity-suppressed by $(m_\mu/m_K)^2$ relative to $K^+ \to \pi^+\pi^0$.

(a) $\sigma(\rho^0)/\sigma(K^0)$ at $E = 2 \times 249\,\text{MeV} = 498\,\text{MeV}$

Use the Breit-Wigner with $E_{\text{cm}} = 498\,\text{MeV} = m_K c^2$.

For K^0 ($J = 0$, on resonance, $B_{\text{in}} = B_{\text{out}} = 0.69$, $m_K = 498$, $\Gamma_K = 7.4 \times 10^{-12}\,\text{MeV}$):

$$\sigma_K \propto (2 \cdot 0 + 1)\,\frac{(0.69)^2\,\Gamma_K^2}{0 + m_K^2\,\Gamma_K^2}.$$

Cancel Γ_K^2:

$$\sigma_K \propto \frac{(0.69)^2}{m_K^2} = \frac{0.4761}{498^2}\,\text{MeV}^{-2}.$$

Evaluate:

$$\sigma_K \propto \frac{0.4761}{2.480 \times 10^5} = 1.920 \times 10^{-6}\,\text{MeV}^{-2}.$$

For ρ^0 ($J = 1$, $B_{\text{in}}B_{\text{out}} = 1$, $m_\rho = 769$, $\Gamma_\rho = 147\,\text{MeV}$, $E = 498$):

$$(E^2 - m_\rho^2)^2 = (498^2 - 769^2)^2\,\text{MeV}^4.$$

Insert squares:

$$(E^2 - m_\rho^2)^2 = (2.480 \times 10^5 - 5.914 \times 10^5)^2 = (3.434 \times 10^5)^2.$$

Evaluate:

$$(E^2 - m_\rho^2)^2 = 1.179 \times 10^{11}\,\text{MeV}^4.$$

And:

$$E^2\Gamma_\rho^2 = (498)^2(147)^2 = 2.480 \times 10^5 \times 2.161 \times 10^4 = 5.359 \times 10^9\,\text{MeV}^4.$$

Sum:

$$D_\rho = 1.179 \times 10^{11} + 5.359 \times 10^9 = 1.233 \times 10^{11}\,\text{MeV}^4.$$

Numerator:

$$N_\rho = 3 \times 1 \times \Gamma_\rho^2 = 3 \times 2.161 \times 10^4 = 6.483 \times 10^4\,\text{MeV}^2.$$

Hence:

$$\sigma_\rho \propto \frac{6.483 \times 10^4}{1.233 \times 10^{11}} = 5.258 \times 10^{-7}\,\text{MeV}^{-2}.$$

Take the ratio:

$$\frac{\sigma_\rho}{\sigma_K} = \frac{5.258 \times 10^{-7}}{1.920 \times 10^{-6}}.$$

Evaluate:

$$\boxed{\left.\frac{\sigma_\rho}{\sigma_K}\right|_{498\,\text{MeV}} \approx 0.27}$$

(b) Beam energy raised to $250\,\text{MeV}$ ($E = 500\,\text{MeV}$)

Now K^0 is just off resonance; redo the denominator for K:

$$(E^2 - m_K^2)^2 = (500^2 - 498^2)^2 = (1.996 \times 10^3)^2 = 3.984 \times 10^6\,\text{MeV}^4.$$

And:

$$E^2\Gamma_K^2 = 2.500 \times 10^5 \times (7.4 \times 10^{-12})^2 = 1.369 \times 10^{-17}\,\text{MeV}^4.$$

The width-squared term is negligible:

$$D_K \approx 3.984 \times 10^6\,\text{MeV}^4.$$

Numerator $N_K = 1 \times 0.4761 \times \Gamma_K^2 = 0.4761 \times 5.476 \times 10^{-23}\,\text{MeV}^4 = 2.607 \times 10^{-23}\,\text{MeV}^4$:

$$\sigma_K \propto \frac{2.607 \times 10^{-23}}{3.984 \times 10^6} = 6.54 \times 10^{-30}\,\text{MeV}^{-2}.$$

For ρ^0 at $E = 500$:

$$(E^2 - m_\rho^2)^2 = (2.500 \times 10^5 - 5.914 \times 10^5)^2 = (3.414 \times 10^5)^2 = 1.165 \times 10^{11}\,\text{MeV}^4.$$

$E^2\Gamma_\rho^2 = 2.500 \times 10^5 \times 2.161 \times 10^4 = 5.402 \times 10^9$. Sum:

$$D_\rho = 1.165 \times 10^{11} + 5.402 \times 10^9 = 1.219 \times 10^{11}\,\text{MeV}^4.$$

$$\sigma_\rho \propto \frac{6.483\times 10^4}{1.219\times 10^{11}} = 5.319\times 10^{-7}\,\mathrm{MeV}^{-2}.$$

Ratio:

$$\left.\frac{\sigma_\rho}{\sigma_K}\right|_{500\,\mathrm{MeV}} = \frac{5.319\times 10^{-7}}{6.54\times 10^{-30}}.$$

Evaluate:

$$\boxed{\left.\frac{\sigma_\rho}{\sigma_K}\right|_{500\,\mathrm{MeV}} \approx 8\times 10^{22}}$$

A 2 MeV shift off the K^0 peak collapses the cross-section by $\sim$ 23 orders of magnitude: $\Gamma_K \sim 10^{-12}$ MeV requires energy resolution $\Delta E \lesssim \Gamma_K \sim 10^{-12}$ MeV, far beyond any beam, so resonant K^0 production via $\pi^+\pi^-$ is impractical.

(c) Radius of curvature of the pion

Pion mass $m_\pi = 139.57\,\mathrm{MeV}/c^2$, $E_\pi = 249\,\mathrm{MeV}$:

$$(p_\pi c)^2 = E^2 - (m_\pi c^2)^2 = 249^2 - 139.57^2.$$

Evaluate squares:

$$(p_\pi c)^2 = 62001 - 19480 = 42521\,\mathrm{MeV}^2.$$

Take square root:

$$p_\pi c = 206.2\,\mathrm{MeV}.$$

Radius of curvature, $r = p/(qB)$, i.e. $r = (pc)/(qBc)$ with $qB = 1\,\mathrm{eV\,s\,m}^{-2}$:

$$r = \frac{2.062\times 10^8\,\mathrm{eV}}{(1\,\mathrm{eV\,s\,m}^{-2})(3\times 10^8\,\mathrm{m\,s}^{-1})}.$$

Evaluate:

$$\boxed{r \approx 0.69\,\mathrm{m}}$$

Question 3 — Fusion, fission and moderation

Problem. pp-fusion start; total H consumed by the Sun and remaining lifetime; SEMF term-by-term; energy released and uranium burned in ^{235}U fission; collisions to thermalise neutrons in graphite.

(a) Initial pp-fusion step

$$\boxed{p + p \longrightarrow {}^2_1H + e^+ + \nu_e, \qquad Q \approx 0.42\,\mathrm{MeV}}$$

(b) Hydrogen burned to date and time remaining

Energy delivered to the Sun's photosphere per chain (annihilation included, neutrinos lost):

$$E_{\text{chain}} = Q + 2\times 1.02\,\mathrm{MeV} - 2\times 0.26\,\mathrm{MeV}.$$

Evaluate:

$$E_{\text{chain}} = 24.68 + 2.04 - 0.52 = 26.20\,\mathrm{MeV}.$$

Per H nucleus consumed (4 H per chain):

$$E_H = E_{\text{chain}}/4 = 6.55\,\text{MeV}.$$

Time elapsed:

$$t = 4.6 \times 10^9\,\text{yr} \times 3.156 \times 10^7\,\text{s}\,\text{yr}^{-1} = 1.452 \times 10^{17}\,\text{s}.$$

Total energy emitted:

$$E_{\text{tot}} = L_\odot\, t = 3.86 \times 10^{26}\,\text{W} \times 1.452 \times 10^{17}\,\text{s}.$$

Evaluate:

$$E_{\text{tot}} = 5.604 \times 10^{43}\,\text{J}.$$

Convert to MeV ($1\,\text{MeV} = 1.602 \times 10^{-13}\,\text{J}$):

$$E_{\text{tot}} = \frac{5.604 \times 10^{43}}{1.602 \times 10^{-13}} = 3.499 \times 10^{56}\,\text{MeV}.$$

Number of H burned:

$$N_H^{\text{burned}} = \frac{E_{\text{tot}}}{E_H} = \frac{3.499 \times 10^{56}}{6.55}.$$

Evaluate:

$$\boxed{N_H^{\text{burned}} \approx 5.34 \times 10^{55}}$$

Remaining H: $9 \times 10^{56} - 5.34 \times 10^{55} = 8.47 \times 10^{56}$. Time remaining at constant luminosity:

$$t_{\text{rem}} = \frac{N_H^{\text{rem}}}{N_H^{\text{burned}}}\, t = \frac{8.47 \times 10^{56}}{5.34 \times 10^{55}} \times 4.6 \times 10^9\,\text{yr}.$$

Evaluate:

$$\boxed{t_{\text{rem}} \approx 7.3 \times 10^{10}\,\text{yr}}$$

(c) SEMF terms

- $Z(m_p + m_e)$: rest mass of Z protons plus their atomic electrons (atomic-mass convention).
- $(A - Z)m_n$: rest mass of $N = A - Z$ neutrons.
- $-a_v A$ (**volume**, attractive): each nucleon feels short-range strong attraction from a roughly fixed number of neighbours; binding energy is therefore extensive in A.
- $+a_s A^{2/3}$ (**surface**, reduces binding): surface nucleons have fewer neighbours; the surface term subtracts the over-counting in the volume term, scaling as the area.
- $+a_c Z^2/A^{1/3}$ (**Coulomb**, reduces binding): electrostatic repulsion between Z protons in a sphere of radius $\propto A^{1/3}$ gives self-energy $\propto Z^2/A^{1/3}$.
- $+a_a(Z - A/2)^2/A$ (**asymmetry/symmetry**, reduces binding when $N \neq Z$): Pauli exclusion + isospin symmetry of the strong force favour $N = Z$; converting protons to neutrons (or vice versa) costs Fermi-level kinetic energy.
- $\pm a_p A^{-1/2}$ (**pairing**): like-nucleon pairs couple to $J = 0$ for extra binding; sign $+$ for odd-odd (less bound), 0 for odd-A, $-$ for even-even (more bound).

(d) Energy released per ^{235}U fission and annual mass burned

Reaction: $n + {}^{235}_{92}\mathrm{U} \to {}^{92}_{37}\mathrm{Rb} + {}^{140}_{55}\mathrm{Cs} + 4n$. Charge $(92 = 37 + 55)$ and baryon $(235 + 1 = 92 + 140 + 4)$ conserved. The mass-difference reduces (as derived above) to

$$Q = B(^{92}\mathrm{Rb}) + B(^{140}\mathrm{Cs}) - B(^{235}\mathrm{U}),$$

with binding energy $B(Z, A) = a_v A - a_s A^{2/3} - a_c Z^2/A^{1/3} - a_a (Z - A/2)^2/A \mp a_p A^{-1/2}$.

^{235}**U** ($Z = 92$, $A = 235$, odd-A so pairing $= 0$):

$$a_v A = 15.56 \times 235 = 3656.6\,\mathrm{MeV}.$$

$$A^{2/3} = 235^{2/3} = 38.08; \quad a_s A^{2/3} = 17.23 \times 38.08 = 656.1\,\mathrm{MeV}.$$

$$A^{1/3} = 6.171; \quad a_c Z^2/A^{1/3} = 0.697 \times 8464/6.171 = 956.0\,\mathrm{MeV}.$$

$$(Z - A/2)^2 = (92 - 117.5)^2 = 649.25; \quad a_a(\cdot)/A = 93.14 \times 649.25/235 = 257.3\,\mathrm{MeV}.$$

Sum:

$$B(^{235}\mathrm{U}) = 3656.6 - 656.1 - 956.0 - 257.3 = 1787.2\,\mathrm{MeV}.$$

^{92}**Rb** ($Z = 37$, $A = 92$, $N = 55$, odd-odd $\Rightarrow$ pairing $+12/\sqrt{A}$):

$$a_v A = 15.56 \times 92 = 1431.5\,\mathrm{MeV}.$$

$$A^{2/3} = 92^{2/3} = 20.38; \quad a_s A^{2/3} = 17.23 \times 20.38 = 351.1\,\mathrm{MeV}.$$

$$A^{1/3} = 4.514; \quad a_c Z^2/A^{1/3} = 0.697 \times 1369/4.514 = 211.4\,\mathrm{MeV}.$$

$$(Z - A/2)^2 = (37 - 46)^2 = 81; \quad a_a \cdot 81/92 = 93.14 \times 0.880 = 82.0\,\mathrm{MeV}.$$

$$a_p A^{-1/2} = 12/\sqrt{92} = 1.25\,\mathrm{MeV} \quad \text{(odd-odd: subtract)}.$$

Sum:

$$B(^{92}\mathrm{Rb}) = 1431.5 - 351.1 - 211.4 - 82.0 - 1.25 = 785.7\,\mathrm{MeV}.$$

^{140}**Cs** ($Z = 55$, $A = 140$, $N = 85$, odd-odd):

$$a_v A = 15.56 \times 140 = 2178.4\,\mathrm{MeV}.$$

$$A^{2/3} = 140^{2/3} = 26.96; \quad a_s A^{2/3} = 17.23 \times 26.96 = 464.5\,\mathrm{MeV}.$$

$$A^{1/3} = 5.192; \quad a_c Z^2/A^{1/3} = 0.697 \times 3025/5.192 = 406.1\,\mathrm{MeV}.$$

$$(Z - A/2)^2 = (55 - 70)^2 = 225; \quad a_a \cdot 225/140 = 93.14 \times 1.607 = 149.7\,\mathrm{MeV}.$$

$$a_p A^{-1/2} = 12/\sqrt{140} = 1.01\,\mathrm{MeV}.$$

Sum:

$$B(^{140}\mathrm{Cs}) = 2178.4 - 464.5 - 406.1 - 149.7 - 1.01 = 1157.1\,\mathrm{MeV}.$$

Combine:

$$Q = 785.7 + 1157.1 - 1787.2.$$

Evaluate:

$$\boxed{Q \approx 156\,\mathrm{MeV}}$$

Per fission in joules:

$$Q = 155.6\,\mathrm{MeV} \times 1.602 \times 10^{-13}\,\mathrm{J/MeV} = 2.493 \times 10^{-11}\,\mathrm{J}.$$

Fission rate at $P = 100\,\mathrm{MW} = 10^8\,\mathrm{W}$:

$$\dot{N} = \frac{P}{Q} = \frac{10^8}{2.493 \times 10^{-11}} = 4.01 \times 10^{18}\,\mathrm{s}^{-1}.$$

Per year ($3.156 \times 10^7\,\mathrm{s}$):

$$N_{\mathrm{yr}} = 4.01 \times 10^{18} \times 3.156 \times 10^7.$$

Evaluate:

$$\boxed{N_{\mathrm{yr}} \approx 1.27 \times 10^{26}\,\mathrm{atoms\,yr}^{-1}}$$

Mass per atom $= 235\,\mathrm{u} \times 1.661 \times 10^{-27}\,\mathrm{kg} = 3.903 \times 10^{-25}\,\mathrm{kg}$:

$$m_{\mathrm{yr}} = 1.266 \times 10^{26} \times 3.903 \times 10^{-25}\,\mathrm{kg}.$$

Evaluate:

$$\boxed{m_{\mathrm{yr}} \approx 49.4\,\mathrm{kg} = 4.94 \times 10^4\,\mathrm{g}}$$

(e) Number of elastic collisions to thermalise

Per collision energy ratio:

$$f \equiv \frac{E_f}{E_i} = \frac{m_C^2 + m_n^2}{(m_C + m_n)^2}.$$

Insert $m_n = 0.94\,\mathrm{GeV}$, $m_C = 11.18\,\mathrm{GeV}$:

$$f = \frac{(11.18)^2 + (0.94)^2}{(12.12)^2} = \frac{124.99 + 0.884}{146.89}.$$

Evaluate:

$$f = \frac{125.87}{146.89} = 0.857.$$

After N collisions $E_N = E_0\, f^N$. Required reduction:

$$\frac{E_N}{E_0} = \frac{0.025\,\mathrm{eV}}{2 \times 10^6\,\mathrm{eV}} = 1.25 \times 10^{-8}.$$

Solve for N:

$$N = \frac{\ln\!\left(1.25 \times 10^{-8}\right)}{\ln 0.857} = \frac{-18.20}{-0.1542}.$$

Evaluate:

$$\boxed{N \approx 118}$$

Question 4 — W and Z widths and the fourth generation

Problem. List W, Z first-generation decays; compute lepton-pair and quark-pair branching ratios; deduce the number of fermion pairs in the W width; mass constraints on a fourth generation; relate the W width to the W/Z cross-section ratio; predict Γ_W, Γ_Z and the cross-section ratio with a fourth generation contributing; compare with CDF and LEP precision.

(a) First-generation decays and branching ratios

Z **(first generation):** $Z \to e^+e^-,\ \nu_e\bar{\nu}_e,\ u\bar{u},\ d\bar{d}.$

W^+ **(first generation):** $W^+ \to e^+\nu_e,\ u\bar{d}.$

Counting open channels for the full W (t-quark too heavy): three lepton flavours $(e\nu_e, \mu\nu_\mu, \tau\nu_\tau)$ and two quark Cabibbo doublets $(u\bar{d}, c\bar{s})$ each in 3 colours. Equal partial widths:

$$N_{\text{channels}} = 3\ +\ 2 \times 3 = 9.$$

Hence:

$$\boxed{\mathrm{BR}(W \to \ell\nu_\ell) = \tfrac{1}{9} \approx 11.1\%, \qquad \mathrm{BR}(W \to q\bar{q}')_{\text{summed colour}} = \tfrac{3}{9} \approx 33.3\%}$$

(b) Number of distinguishable fermion pairs in Γ_W

Equal partial widths, so

$$N_{\text{pairs}} = \frac{\Gamma_W^{\text{tot}}}{\Gamma(W \to e\nu)} = \frac{2085}{232}.$$

Evaluate:

$$\boxed{N_{\text{pairs}} \approx 9}$$

Consistent with three lepton flavours and two colour-tripled quark doublets; no fourth generation in W decay.

(c) Mass constraints from W, Z widths

The measured widths leave no room for new on-shell decays. Therefore each fourth-generation pair must be kinematically forbidden:

$$\boxed{m_T + m_B > m_W, \qquad m_\sigma + m_N > m_W}$$

$$\boxed{2m_T,\, 2m_B,\, 2m_\sigma,\, 2m_N > m_Z}$$

i.e. each new fermion mass $> m_Z/2 \approx 45.6\,\text{GeV}$, with the additional pair-sum constraints from W decay.

(d) Indirect determination of Γ_W

At the resonance peak ($E = m$), with $J = 1$ for both W and Z, the Breit-Wigner reduces to

$$\sigma(m) = K\,(2J+1)\,\frac{B_{\text{in}}B_{\text{out}}\,\Gamma_{\text{tot}}^2}{m^2\,\Gamma_{\text{tot}}^2} = 3K\,\frac{\Gamma_{\text{in}}\Gamma_{\text{out}}}{m^2\,\Gamma_{\text{tot}}^2}.$$

Define the measured ratio:

$$R \equiv \frac{\sigma(W \to \ell\nu)}{\sigma(Z \to \ell\ell)}.$$

Substitute the peak expressions (K cancels by assumption):

$$R = \frac{\Gamma_{\text{in}}^W\,\Gamma_{\text{out}}^W}{m_W^2\,\Gamma_W^2} \Bigg/ \frac{\Gamma_{\text{in}}^Z\,\Gamma_{\text{out}}^Z}{m_Z^2\,\Gamma_Z^2}.$$

Rearrange for Γ_W^2:

$$\Gamma_W^2 = \frac{\Gamma_{\text{in}}^W\,\Gamma_{\text{out}}^W\,m_Z^2\,\Gamma_Z^2}{R\,\Gamma_{\text{in}}^Z\,\Gamma_{\text{out}}^Z\,m_W^2}.$$

Take square root:

$$\boxed{\Gamma_W = \frac{m_Z\,\Gamma_Z}{m_W}\sqrt{\frac{\Gamma^W_{\rm in}\,\Gamma^W_{\rm out}}{R\,\Gamma^Z_{\rm in}\,\Gamma^Z_{\rm out}}}}$$

(e) Effect of a fourth generation of leptons on Γ_W, Γ_Z and R

Per-flavour partial widths:

$$\Gamma(Z \to \nu\bar{\nu})|_{\rm flavour} = \frac{501}{3} = 167\,\mathrm{MeV}, \qquad \Gamma(Z \to \ell^+\ell^-)|_{\rm flavour} = 84\,\mathrm{MeV}.$$

Add one fourth generation (one ν_4, one ℓ_4):

$$\Gamma'_Z = 2495 + 167 + 84 = 2746\,\mathrm{MeV}.$$

Boxed:

$$\boxed{\Gamma'_Z = 2746\,\mathrm{MeV}}$$

For W, add one extra lepton-pair channel of partial width $232\,\mathrm{MeV}$:

$$\Gamma'_W = 2085 + 232 = 2317\,\mathrm{MeV}.$$

Boxed:

$$\boxed{\Gamma'_W = 2317\,\mathrm{MeV}}$$

Ratio sensitivity. From (d), at fixed standard-model partial widths,

$$R \;\propto\; \frac{1/\Gamma_W^2}{1/\Gamma_Z^2} = \frac{\Gamma_Z^2}{\Gamma_W^2}.$$

So:

$$\frac{R'}{R} = \left(\frac{\Gamma'_Z}{\Gamma_Z}\right)^2 \Big/ \left(\frac{\Gamma'_W}{\Gamma_W}\right)^2.$$

Insert numbers:

$$\frac{\Gamma'_Z}{\Gamma_Z} = \frac{2746}{2495} = 1.1006; \qquad \frac{\Gamma'_W}{\Gamma_W} = \frac{2317}{2085} = 1.1113.$$

Square and divide:

$$\frac{R'}{R} = \frac{(1.1006)^2}{(1.1113)^2} = \frac{1.2114}{1.2350} = 0.9809.$$

Fractional change:

$$\boxed{\frac{\Delta R}{R} = -1.91\%}$$

(f) Significance at CDF and LEP

CDF σ-ratio (1.9% precision):

$$n_\sigma^{\rm CDF} = \frac{|\Delta R/R|}{0.019} = \frac{0.0191}{0.019}.$$

Evaluate:

$$\boxed{n_\sigma^{\rm CDF} \approx 1.0\,\sigma}$$

LEP Γ_Z (2.3 MeV precision):

$$n_\sigma^{\mathrm{LEP}} = \frac{\Delta\Gamma_Z}{2.3\,\mathrm{MeV}} = \frac{2746 - 2495}{2.3} = \frac{251}{2.3}.$$

Evaluate:

$$\boxed{n_\sigma^{\mathrm{LEP}} \approx 109\,\sigma}$$

LEP rules out a low-mass fourth generation overwhelmingly; CDF, by contrast, has only marginal sensitivity through the cross-section ratio.